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(54) **CLAMSHELL PICKLEBALL PADDLE**

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(57)

ABSTRACT

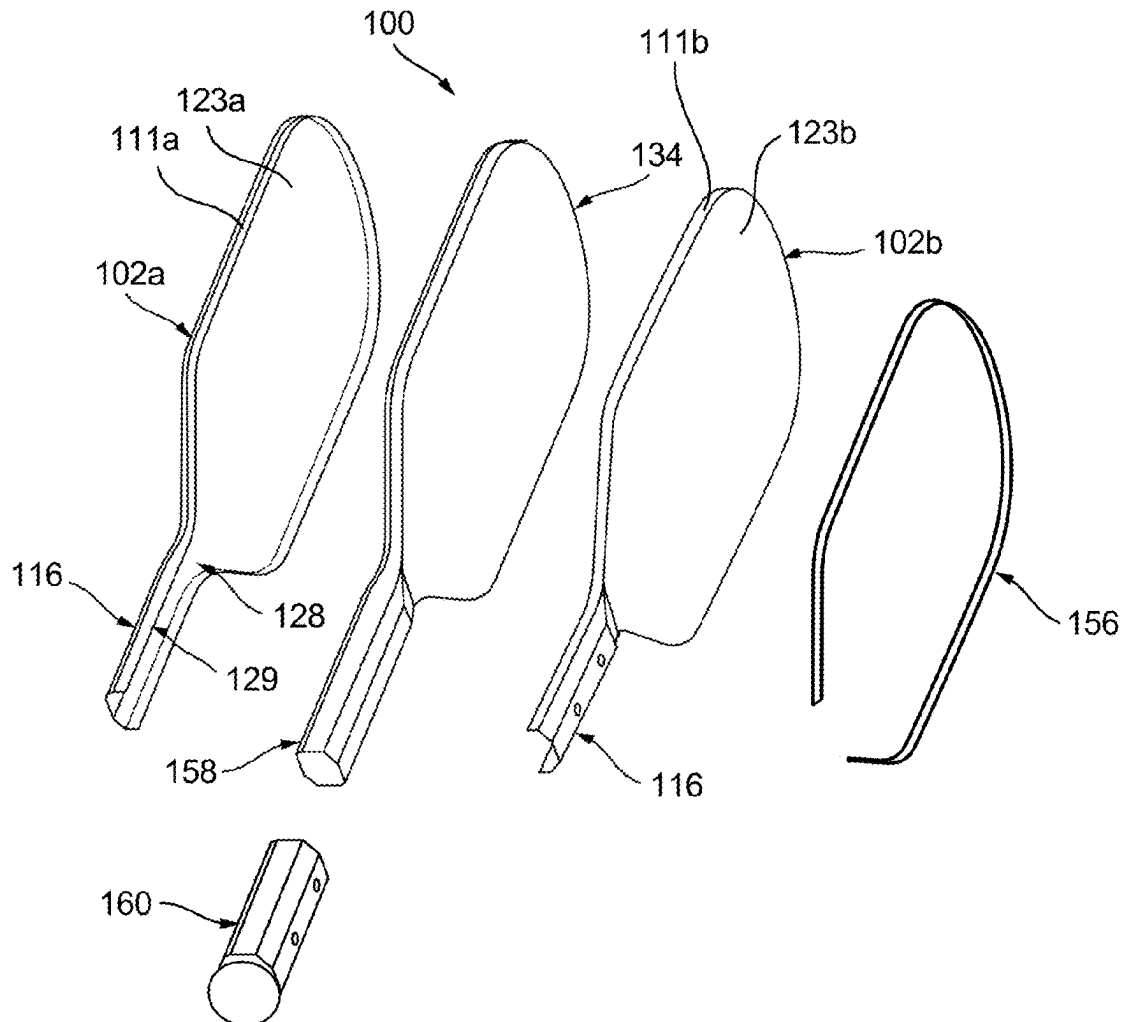
Described herein are various embodiments of a pickleball paddle comprising a first clamshell component and a second clamshell component. The first clamshell component includes a first head portion, a first rim portion surrounding the first head portion, and a first handle portion. The second clamshell component includes a second head portion, a second rim portion surrounding the second head portion, and a second handle portion. When the first rim portion is joined to the second rim portion, the first and second clamshell components together define a cavity. A core is disposed in the cavity and may be formed of a foam material.

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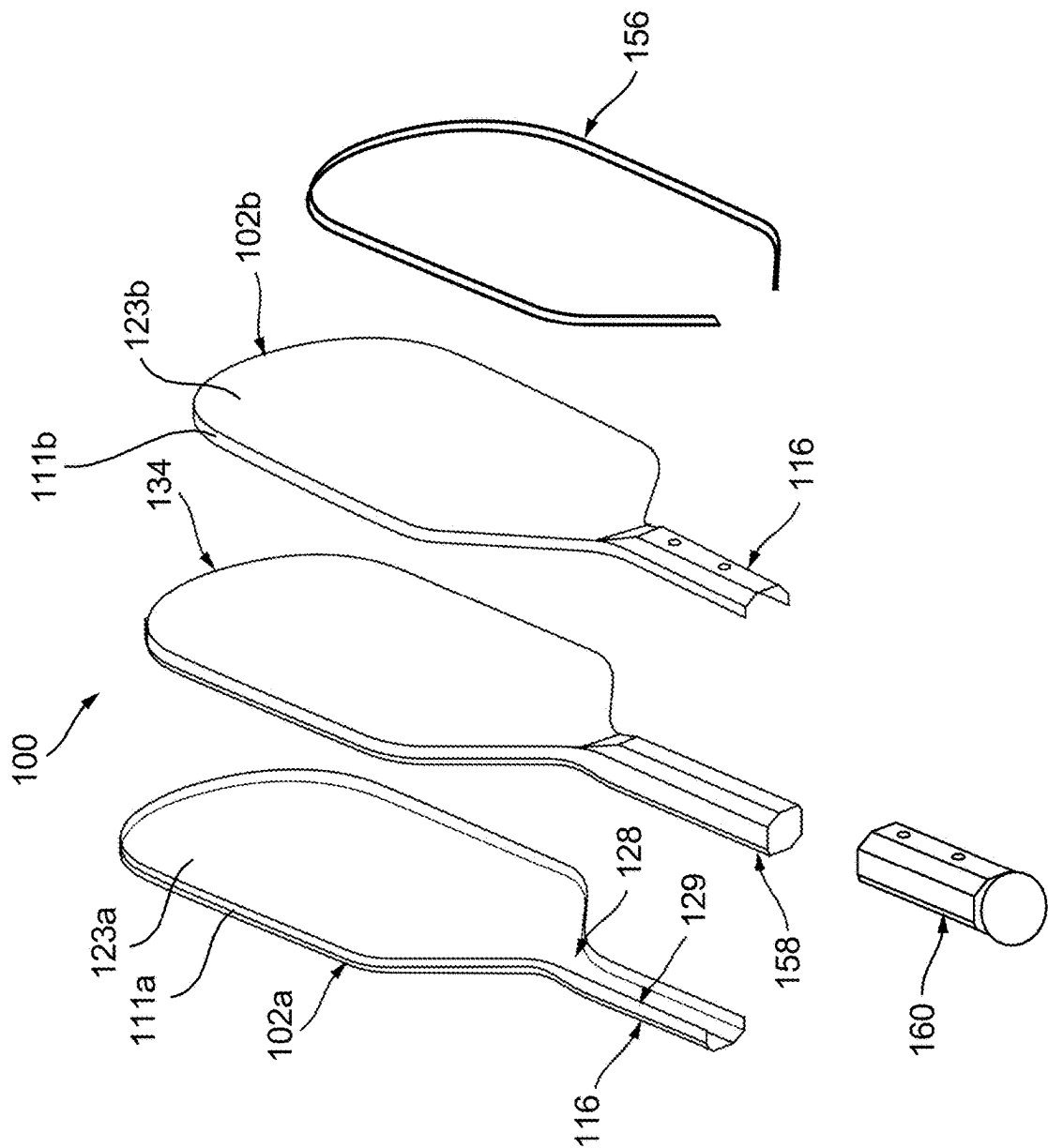


FIG. 1

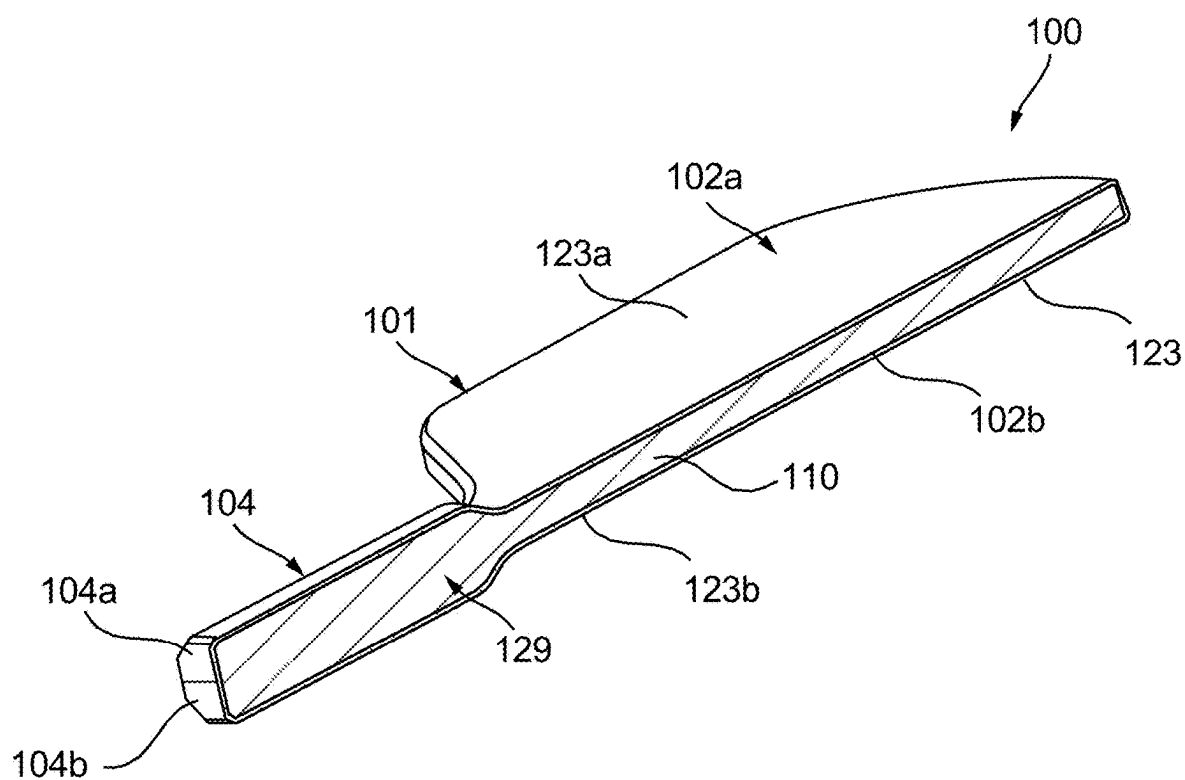


FIG. 2

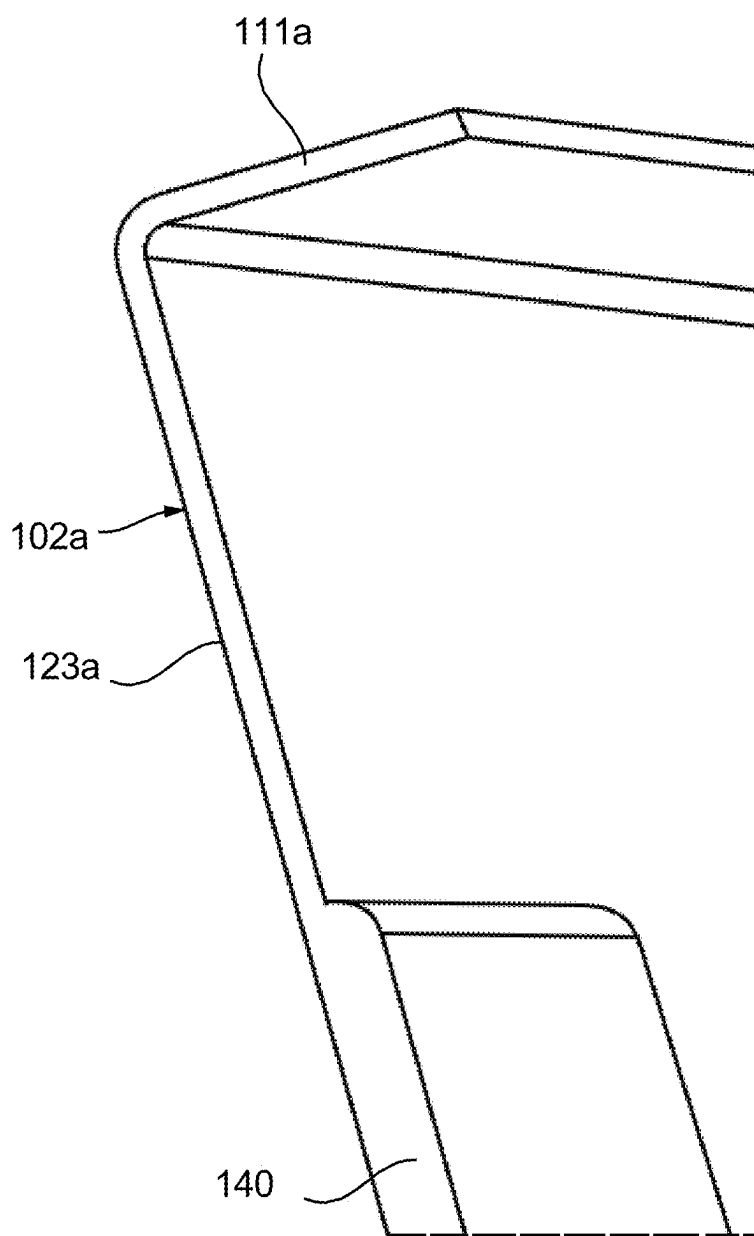


FIG. 3

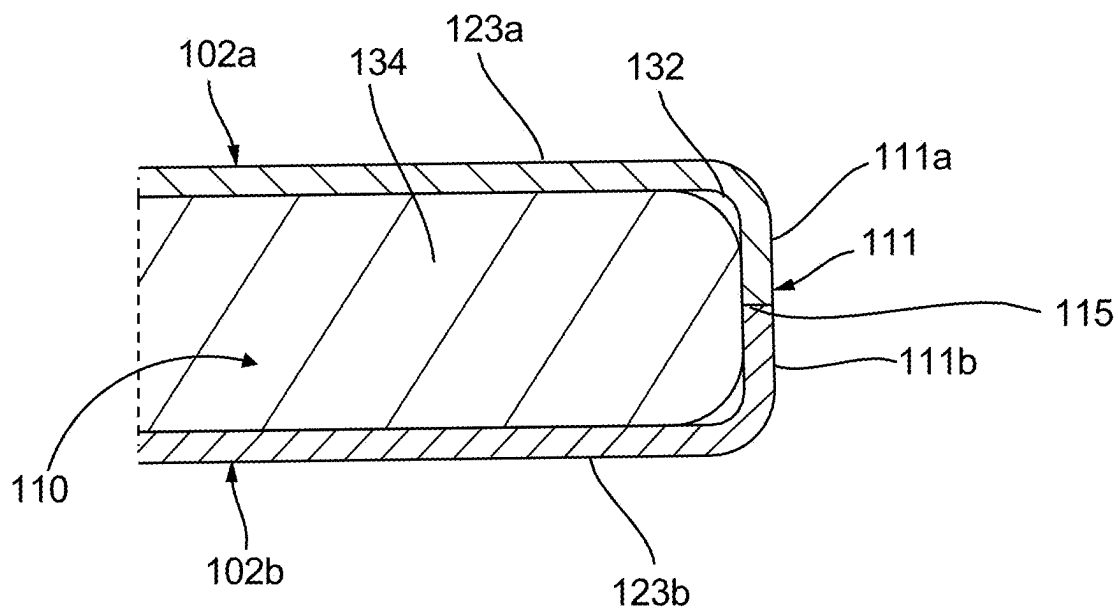


FIG. 4A

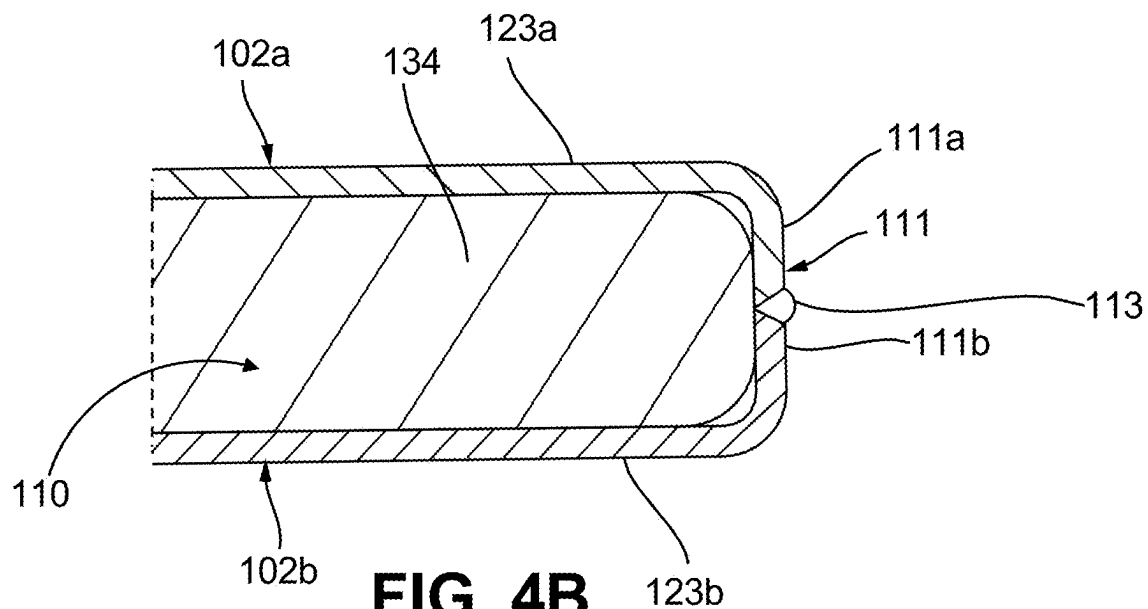


FIG. 4B

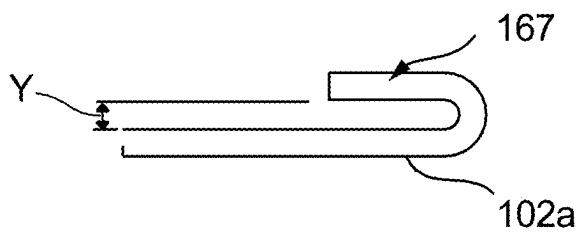


FIG. 5A

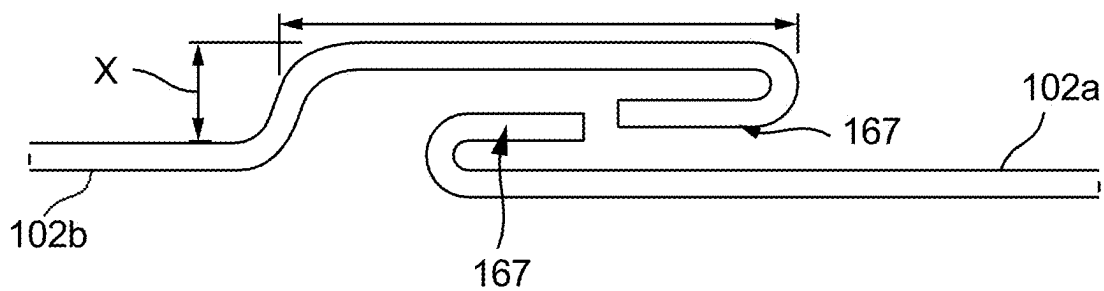


FIG. 5B

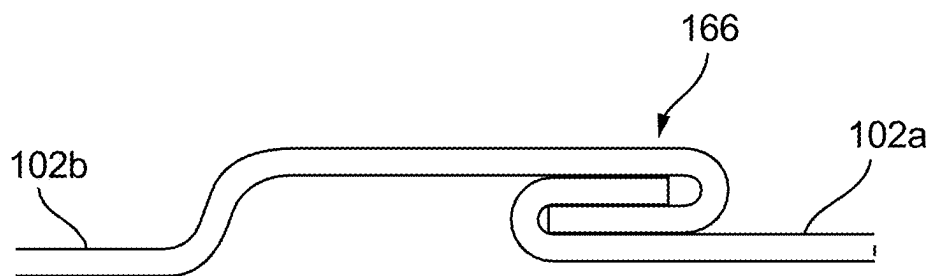


FIG. 5C

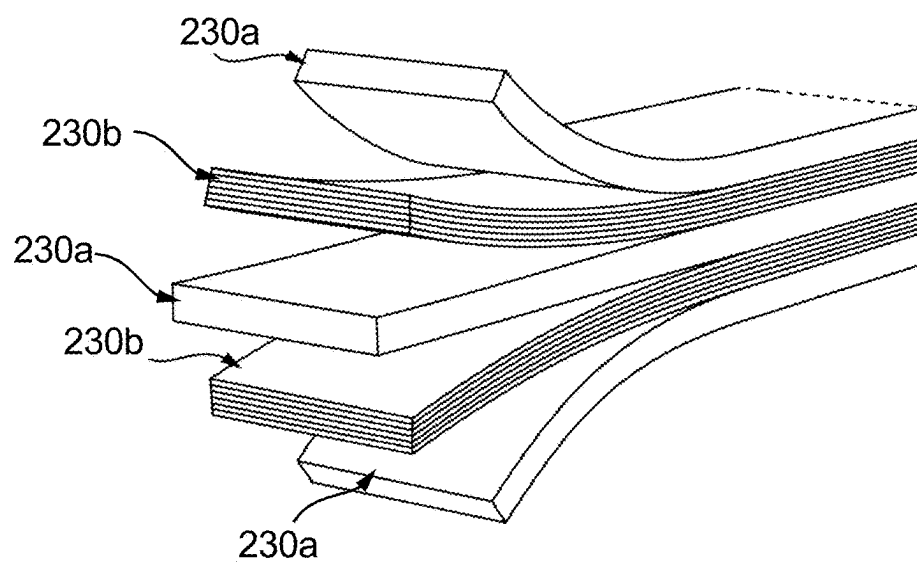


FIG. 6

CLAMSHELL PICKLEBALL PADDLE

TECHNICAL FIELD

[0001] This invention generally relates to pickleball equipment, and more particularly, to pickleball paddles.

BACKGROUND

[0002] Pickleball paddles typically comprise a core that directly supports skins or face plates. Conventional face plates are attached to or integrally formed with a handle and surrounded by an edge guard to cover and protect the periphery of the core. The process of forming each individual part and assembling them together is lengthy and can be complicated, while leaving little room for customization. Additionally, the core of traditional paddles is the primary structural component which limits the type of materials that can be used for the core. Consequently, conventional pickleball paddles lack manufacturability and modularity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] To facilitate further description of the embodiments, the following drawings are provided in which:

[0004] FIG. 1 is an exploded perspective view of a pickleball paddle, according to an embodiment.

[0005] FIG. 2 is a perspective view, in cross-section, of a pickleball paddle.

[0006] FIG. 3 is an enlarged perspective view of a clamshell section of the pickleball paddle of FIG. 1.

[0007] FIG. 4A is an enlarged side elevation view, in cross-section, of a portion of the pickleball paddle.

[0008] FIG. 4B is an enlarged side elevation view, in cross-section, of a portion of the pickleball paddle of FIG. 1, with clamshell sections joined by welding.

[0009] FIG. 5A is an enlarged view of one portion of a hem joint.

[0010] FIG. 5B is an enlarged, exploded view of a hem joint.

[0011] FIG. 5C is an enlarged view of a hem joint coupling portions of a pickleball paddle.

[0012] FIG. 6 is an enlarged perspective view showing layers of material used to form the clamshell.

DEFINITIONS

[0013] For simplicity and clarity of illustration, the drawing figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the invention. Additionally, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of embodiments of the present invention. The same reference numerals in different figures denote the same elements.

[0014] The terms “first,” “second,” “third,” “fourth,” and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein.

[0015] Furthermore, the terms “include,” and “have,” and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, system, article, device, or apparatus that comprises a list of elements is not necessarily limited to those elements but may include other elements not expressly listed or inherent to such process, method, system, article, device, or apparatus.

[0016] The terms “front,” “back,” “top,” “bottom,” “over,” “under,” “north,” “south,” “east,” “west,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

[0017] The terms “couple,” “coupled,” “couples,” “coupling,” and the like should be broadly understood and refer to connecting two or more elements or signals, electrically, mechanically and/or otherwise.

[0018] The term “geometric centerpoint,” or “geometric center” of the face plate, as used herein, can refer to a geometric centerpoint of the face plate perimeter, and at a midpoint of the face height of the face plate. In the same or other examples, the geometric centerpoint also can be centered with respect to an engineered impact zone, which can be defined by a region of grooves on the face plate.

[0019] The “length” of the pickleball paddle, as described herein, can be defined as a top-to-bottom dimension of the pickleball paddle. In many embodiments, the length of the paddle can be measured according to and approved by a pickleball governing body such as USA PICKLEBALL.

[0020] The term “core” of the pickleball paddle, as described herein, refers to a component that fills at least a portion of the cavity defined by the head and/or handle of the paddle. The core is positioned between the face plates and provides stiffness and support to the face plates.

[0021] The term “head” of the pickleball paddle, as described herein, refers to the large upper portion of the paddle, from a top edge to a handle. The head comprises a rim, a hitting surface, and a panel subassembly.

[0022] The term “rim” of the pickleball paddle, as described herein, refers to a structural material that extends around the periphery of the head, surrounding a majority of the panel subassembly and face plate circumference.

[0023] The term “handle” of the pickleball paddle, as described herein, refers to a portion of the paddle attached to the head, which is configured to be gripped by the user. The handle can include a grip covering for comfort and interchangeability.

DESCRIPTION

[0024] Described herein is a pickleball paddle having a modular clamshell design that facilitates changing of components, simplifies manufacturing, and expands customizing of parts. The clamshell paddle 100 comprises two exterior structural components 102a, 102b that are joined together to form the external surface of the pickleball paddle and define an entirely enclosed cavity 110. In many embodiments, the clamshell components 102a, 102b are identical. In other embodiments, the clamshell components 102a, 102b are not identical but have complementary geometries that can be joined to enclose the cavity 110.

[0025] The clamshell paddle 100 includes a head 101, having face plates 123a, 123b on opposite sides and a handle 104 to be gripped by a user. In the illustrated embodiments, the head 101 and the handle 104 are integrally formed by the clamshell components 102a, 102b, which provide structural support and form a hollow chassis that can house internal components, such as a core 134. The clamshell construction described herein is easier to manufacture and customize and has discretionary space for additional components, while expanding the types of materials that can be used for the core.

A. Peripheral Rim

[0026] The head 101 formed by clamshell components 102a, 102b defines face plates 123a, 123b and a portion of the peripheral rim 111 extending around the face plates 123a, 123b. Each of the clamshell components 102a, 102b integrally form a portion of the rim 111 and one of the face plates 123a, 123b. More specifically, each clamshell component 102a, 102b includes a side wall 223a, 223b, and together the side walls 223a, 223b form the rim 111. Still further, each clamshell component 102a, 102b may include integral handle portion 104a, 104b. The clamshell paddle 100 does not comprise a structural frame separate from the clamshell components 102a, 102b.

[0027] The clamshell components 102a, 102b have side walls 111a, 111b extending perpendicular to the face plate 122 that each partially enclose a side edge of the cavity 110. The side walls 111a, 111b surround the entire face plates 123a, 123b, except for the portions of the face plates 123a, 123b integrally joined with the handle 104. A first side wall 111a of a first clamshell component 102a extends toward and abuts a second side wall 111b of the second clamshell component 102b. In some embodiments, the first side wall 111a and the second side wall 111b are completely aligned with one another. In other embodiments, the side walls 111a, 111b are offset from one another along at least a portion of the peripheral rim 111. In an alternate embodiment, only the first clamshell component 102a comprises a side wall 111a, while the second clamshell component 102b lacks a side wall, such that, when assembled, the first side wall 111a abuts the second face plate 123b. The offset can allow for the incorporation of additional fastening means. In some examples a hem joint 166, as shown in FIGS. 5A-5C, or a lacing system can be used to join together and fasten the two clamshell components 202a, 202b. In many embodiments, each clamshell component 202a, 202b includes a handle portion 104a, 104b that is later covered by a grip 260 that may help secure together the handle portion 104a, 104b. In some embodiments, grip tape or another elastic grip material is applied over the grip.

[0028] Referring to FIG. 1, in many embodiments, the clamshell paddle 200 comprises a perimeter guard or edge guard 156 to prevent damage, add weight, provide a smoother surface, and/or improve durability and aesthetics. In other embodiments, the clamshell paddle 200 does not comprise a perimeter guard 156. In these embodiments, a coating or wrap may be applied to the peripheral rim 111 to hide the seam 115 between the clamshell components 102a, 102b.

B. Core

[0029] The cavity 110 may be partially or entirely filled by the core 134, as best shown in FIGS. 1 and 2. The core 134

may provide structural rigidity to the paddle, and may be formed of one or more core materials that produce a desired impact response to a ball. In some embodiments, the core facilitates securing together the clamshell components. In these embodiments, the two clamshell components 102a, 102b can cover and enclose the core 134. In some embodiments, other internal components or structures in addition to the core 134 may be disposed in the cavity 110, including but not limited to: a filler material(s), damping layers, supports, and/or adhesives or mechanical fasteners. In some embodiments, as best shown in FIG. 2, the core 134 may include an adhesive layer for attaching to the face plates 123a, 123b.

[0030] The core 134 supports the face plate 123a, 123b and can be configured to modulate characteristics such as sound, feel, and energy transfer. In many embodiments, the core 134 occupies the majority or entirety of the cavity 110. In some embodiments, the first clamshell section 202a and the second clamshell section 202b define the cavity 110. In other embodiments, the first face plate 123a, the second face plate 123b, and the frame 102 define the cavity 110. In some embodiments, the core 134 further occupies the majority or entirety of the handle cavity 129. In many embodiments, the core 134 occupies the majority or entirety of the cavity 110, but only a portion of the handle cavity 129. In some of these embodiments, handle fill material can be added to occupy a remainder of the handle cavity 129. In many embodiments, the core 134 includes multiple core layers and/or regions formed of different core structural materials. For example, a first core structural material and a second core structural material may occupy the cavity 110, while a third core structural material may occupy the handle cavity 129.

[0031] In some embodiments, the core 134 itself can be an injectable core material. The core material can be selected to attenuate sound and distribute weight. A lightweight core material may be used to balance out heavy materials used for the clamshell components 102a, 102b. Alternatively, a heavier core material may be used to provide support and additional mass when a lightweight material is used for the frame 102 and/or face plates 123a, 123b, in other embodiments.

[0032] In some embodiments, the core material may be added to fill most or all of the cavity 110. For example, the core material can be an expanding foam material selected to bond with the material forming the face plates 123a, 123b. In some embodiments, the expanding foam is used in addition to at least one other core material.

[0033] The core 134 can comprise a core thickness. The core thickness can be between 0.01 inch and 0.99 inch. In many embodiments, the core thickness is greater than the thicknesses of the face plates 123a, 123b. In some embodiments, the core thickness is generally consistent throughout. In other embodiments, the core thickness varies to provide additional mass or support to particular regions throughout paddle 100.

C. Handle

[0034] The handle 104 provides a comfortable means of gripping the paddle and can be used to enhance rigidity of the paddle 100. The handle 104, together with the head 101, externally supports the core 134 and forms the primary structural support for the paddle 100. Integrally forming the handle and the head better distributes stresses between these components and reduces risk of failure. As mentioned previously, and shown in FIGS. 1 and 2, each clamshell

component **102a**, **102b** comprises a handle portion **104a**, **104b** which couple together to form handle **104**. The handle components **104a**, **104b** are integrally formed with each respective clamshell component **102a**, **102b**.

[0035] The handle **104** is positioned proximal to a lower end of the head **101**. The handle **104** has an exterior surface and defines a handle cavity **129** that may fluidly communicate with the cavity **210**. The handle cavity **129** can be unfilled and hollow or filled with a handle fill material. In some embodiments, the handle fill material is contiguous with at least a portion of the core **134**. In other embodiments, the handle fill material comprises a different material than the core **134**, to provide a different feel or weight. In some of these embodiments, the handle fill material can have a greater density than the core material, thereby increasing overall weight of the paddle **100** without greatly increasing the force required to swing the paddle **100**.

[0036] The handle **104** can be covered by a grip **160** intended to be grasped by the player. The grip **160** can comprise one or multiple pieces which slide over the handle **104** as a unit, as best shown in FIG. 1. In other embodiments, the grip can be made of multiple pieces which fasten together over the handle **104**. The grip **160** surrounds the entire circumference of the handle **104**, hiding the seam **115** and providing an additional fastening means, to hold together the clamshell components **102a**, **102b**. The grip can be made of a material which provides cushion and friction to improve the user's comfort and ability to grasp the paddle without the hand slipping.

D. Clamshell Coupling

[0037] The clamshell components **102a**, **102b** are joined directly or indirectly to each other to provide a durable frame for supporting the core. For example, as best shown in FIG. 4A, the clamshell components **102a**, **102b** can be directly coupled with adhesives, mechanical fastening means, or a combination of both. Alternatively, the clamshell components **102a**, **102b** can be joined by an intermediate component, such as a weld **113**, as best shown in FIG. 4B. In other embodiments, the intermediate component can be an epoxy, tape, or preformed fastening material. As stated above, the core structural or filler material can provide adhesion that couples the clamshell components **102a**, **102b**. Alternatively, an adhesive can be used to adhere both clamshell components **102a**, **102b** to the core, thereby indirectly coupling the clamshell components **102a**, **102b**.

[0038] The core **134** can comprise a structural material that is any one or combination of the following materials: wood, such as balsa or bamboo, polypropylene foam, PVC foam, cork, ionomer foam, polyurethane foam, polyethylene sheet, hard felt, synthetic felt, polypropylene honeycomb structure, other expanding or injectable foam, or Lantor Soric foam. Some of these core structural materials can comprise an attached adhesive backing to adhere to the clamshell components **102a**, **102b**. Some core structural materials may require an adhesive, such as epoxy or VHB tape, to attach to the clamshell components **102a**, **102b**.

[0039] In many embodiments, the core material can also be used as a fastener, as best shown in FIG. 2. In these embodiments, an expanding or injectable foam can be applied to the clamshell cavity **110**. In some of these embodiments, the foam is applied to one or both interior sides of the hitting surfaces, and can entirely or partially fill the clamshell cavity **110** while also securing the clamshell

components **102a**, **102b** together. In some embodiments, the foam is also applied to the interior surface of the handle portion **104a**, **104b** while, in other embodiment, the handle **104** is left empty or filled with a different material having different properties, such as density and weight. In some of these embodiments, the handle **104** is filled with a higher density material than that of the core material, thereby providing additional weight to the handle **104**. The process or using a single material as both the core **134** and the fastener is uncomplicated and quick, saving time and minimizing risk of application or assembly error.

[0040] The grip **160** can further fasten together the clamshell components. In some embodiments, as shown in FIG. 1, the grip can be a single component which slides over the clamshell handle portion **104**, preventing movement of the handle portion of the clamshell components **104a**, **104b** relative to the other clamshell component handle portion **104a**, **104b** and, therefore, maintaining the clamshell components' **102a**, **102b** coupling.

[0041] In some embodiments, the clamshell components **102a**, **102b** can be fastened, in part, by lacing them to one another. In some of these embodiments, a plurality of apertures can be defined within the peripheral lip of each of the first clamshell component and the second clamshell component. Each aperture in the first clamshell component **102a** can correspond to an aperture in the second clamshell component **102b**. A lace can be inserted through the apertures to secure the clamshell components **102a**, **102b** together. The lace can be a string or cable made of a plastic, metal, or organic material. In some embodiments, the clamshell components **102a**, **102b** are coupled with a lace in addition to adhesive, such as an epoxy or expanding foam.

[0042] The clamshell components **102a**, **102b** can comprise structures that assist in fastening and further prevent separation. In some embodiments, each clamshell components **102a**, **102b** can be bent, molded, or formed to include one side of a hem joint **166** that can be joined together to more securely interlock the clamshell components **102a**, **102b**, as best shown in FIGS. 5A-5C. The hem joint **166** structure of the first clamshell components **102a**, **102b** can correspond with the hem joint **166** structure of the second clamshell component **102b**, such that each hem joint **166** structure can fit together to attach the two clamshell components **102a**, **102b**. In some embodiments, the hem joint **166** can couple the two clamshell components **102a**, **102b**, while they are also coupled by an adhesive, such as an expanding foam.

[0043] The hem joint **166** can resemble a hook **167**, as shown in FIG. 5A. Each clamshell component **102a**, **102b** can comprise a hem joint hook **167**, as shown in FIGS. 5B and 5C. The first clamshell component hem joint hook **167** can be curved in a direction toward the center of the head, while the second clamshell component hem joint hook **167** can be curved in a direction away from the center of the head, or vice versa. The hem joint hooks **167** of each clamshell component **102a**, **102b** must be curved in opposite directions relative to one another in order to interact with one another when assembled.

[0044] One of the hem joint hooks **167** can be positioned a distance X from the side wall, as shown in FIG. 5B. The distance X allows space for said hem joint hook **167** to extend around the opposing hem joint hook **167**, so that the

two hem joint hooks **167** can be interlocked. In some embodiments, the distance X can be between 0.080 and 0.5 inch.

[0045] The hem joint hook can create a gap define by a distance Y between the side wall and the edge of the hem joint hook. The distance Y can be 0.060 inch to 0.250 inch. In some embodiments, the distance Y can be 0.060 inch to 0.080 inch, 0.080 inch to 0.100 inch, 0.100 inch and 0.120 inch, 0.120 inch to 0.168 inch, 0.168 inch to 0.160 inch, 0.160 inch to 0.180 inch, 0.180 inch to 0.200 inch, 0.200 inch to 0.220 inch, 0.220 inch to 0.250 inch. In some examples, the distance Y can be 0.060 inch, 0.090 inch, 0.125 inch, 0.187 inch, or 0.250 inch.

[0046] The hem joint hooks are illustrated with gaps in the shape of curves or arcs, creating a semicircular shape, but can comprise an angular shape, constructed from straight lines. In one example, the hook can comprise a rectangular shape. The hem joint is created by bending the edge of the side wall 90 degrees to greater than 180 degrees, depending on the type of hem joint used.

[0047] The hem joint hooks can extend around a portion of the head portion side walls. The hem joint hooks can extend around 20% to 100% of the head portion side walls. In some examples, the hem joint hooks extend around 20% to 30%, 30% to 40%, 40% to 50%, 50% to 60%, 60% to 70%, 70% to 80%, 80% to 90%, or 90% to 100% of the head portion side walls. The hem joint hooks can be positioned in a single, continuous portion of the periphery of the head, or can be segmented into multiple sections, while sections of the side wall are devoid of hem joint hooks.

[0048] An adhesive, filler, coating material, band, or perimeter guard can be applied to the periphery of the head **101** or the head **101** together with the handle **104**, to provide additional fastening security, to hide the seam **115** where the two clamshell sections **102a**, **102b** meet, and/or to prevent debris from entering the hollow interior defined by the clamshell sections **102a**, **102b**.

E. Variable Hitting Surface Thickness

[0049] The structures and/or materials of the components of the paddle **100** may vary across the face plate **123a**, **123b** to modify the response of the paddle during impact, altering characteristics such as feel, sound, and energy transfer. Additionally, varying structures and/or materials may improve control, consistency, and power. For example, thickness or density of regions of the face plate **122**, can be varied to produce desired performance characteristics.

[0050] One or both of the face plates **123a**, **123b** can comprise variable thickness across its surface. Strategically placed thicker and/or thinner regions can provide a playing surface with a variable surface response to impacts with a ball, creating regions that provide increased or reduced power and control, which are oftentimes trade-offs of one another.

[0051] Integrally formed thickened regions or mounds **140** on the interior of the face plates **123a**, **123b**, as best shown in in FIG. 3, may provide variable thickness. These can be constructed such that the interior surfaces of the face plates **123a**, **123b** vary in height, while the exterior surfaces of the face plates **123a**, **123b** are relatively flat. In additional embodiments, additional layers of material may be located in targeted regions across the face plates **123a**, **123b** to vary the thicknesses of the face plates **123a**, **123b**. In other embodiments, thin regions of the face plates **123a**, **123b** can

be formed by machining, grinding, or other subtractive manufacturing process. In another example, material can be removed from the exterior surface of the face plates **123a**, **123b**, and a coating or exterior layer can be applied to provide a continuous, even surface, as well as desired surface texture.

Dimensions

[0052] The face plates **123a**, **123b** can comprise a face plate length. The face plate length is measured from the lower end—which abuts the handle—to the upper end. In some embodiments, the face plate length can be between 7 inches to 17 inches. In some embodiments, the face plate length can be between 7 inches to 8 inches, 8 inches to 9 inches, 9 inches to 10 inches, 10 inches to 11 inches, 11 inches to 12 inches, 12 inches to 13 inches, 13 inches to 14 inches, 14 inches to 15 inches, 15 inches to 16 inches, or 16 inches to 17 inches. In one exemplary embodiment, the face plate length is 11.75 inches.

[0053] The face plates **123a**, **123b** can comprise a face plate **122** width. The face plate **122** width is measured from the west lateral edge to the east lateral edge. In some embodiments, the face plate **122** width can be between 7 inches to 17 inches. In some embodiments, the face plate **122** width can be between 7 inches to 8 inches, 8 inches to 9 inches, 9 inches to 10 inches, 10 inches to 11 inches, 11 inches to 12 inches, 12 inches to 13 inches, 13 inches to 14 inches, 14 inches to 15 inches, 15 inches to 16 inches, or 16 inches to 17 inches. In one exemplary embodiment, the face plate **122** width is 7.5 inches.

[0054] The face plates **123a**, **123b** can comprise a face plate thickness. In some embodiments, the face plate thickness can be between 0.001 inch and 0.030 inch. In some embodiments the face plate thickness can be between 0.001 inch to 0.005 inch, 0.005 inch to 0.010 inch, 0.010 inch to 0.015 inch, 0.015 inch to 0.020 inch, 0.020 inch to 0.025 inch, or 0.025 inch to 0.030 inch. In specific regions where additional support or stiffening is required, the face plate thickness can be greater than the surrounding face plate thickness. In these regions, the face plate thickness can be greater than 0.010 inch, 0.015 inch, 0.020 inch, 0.025 inch, or 0.030 inch.

[0055] The pickleball paddle **100** can comprise a mass. In some embodiments, the paddle mass can be between 170 grams and 280 grams. In some embodiments, the paddle mass can be between 170 grams and 185 grams, 185 grams and 200 grams, 200 grams and 215 grams, 215 grams and 230 grams, 230 grams and 245 grams, 245 grams and 260 grams, or 260 grams and 280 grams. In one exemplary embodiment, the paddle mass is 215 grams.

I. Methods of Manufacturing and Assembly

A. Materials, Manufacturing, and Assembly

[0056] The clamshell components can be thermoformed, injection molded, pressed, forged, stamped, or otherwise formed. The injection molded or thermoformed pieces can be created out of a molded thermoplastic material. A molded thermoplastic material is one that relies on the polymer itself to provide structure and rigidity to the final component. The molded thermoplastic material is one that is readily adapted to molding techniques where the material is freely flowable when heated to a temperature above the melting point of the

polymer. The thermoplastic material can be a filled thermoplastic (FT) material or an unfilled thermoplastic (UT) material. The thermoplastic material should preferably incorporate one or more engineering polymers that have sufficiently high material strengths and/or strength/weight ratio properties to withstand typical pickleball use while still providing weight savings that are beneficial to the present design. When injection molding, a thermoplastic is heated until melted, then injected into a mold and allowed to cool and cure to take on the shape of the mold. When thermoforming, a thermoplastic sheet is heated until soft and pliable, then stretched and bent into the desired shape by way of vacuum and/or pressing.

[0057] The clamshell components **102a**, **102b** can be made from a wood, plastic, metal, or composite material. In some embodiments, the clamshell components **102a**, **102b** are made of a combination of metal, wood, and carbon fiber composite materials. In thermoformed or injection molded embodiments, the clamshell can be made of any of the following materials: polycarbonate (PC), polyester (PBT), polyphenylene sulfide (PPS), polyamide (PA) (e.g., polyamide 6 (PA6), polyamide 6-6 (PA66), polyamide-12 (PA12), polyamide-612 (PA612), polyamide 11 (PA11)), thermoplastic polyurethane (TPU), polyphthalamide (PPA), acrylonitrile butadiene styrene (ABS), polybutylene terephthalate (PBT), polyvinylidene fluoride (PVDF), polyethylene (PE), polyphenylene ether/oxide (PPE), polyoxymethylene (POM), polypropylene (PP), styrene acrylonitrile (SAN), polymethylpentene (PMP), polyethylene terephthalate (PET), acrylonitrile styrene acrylate (ASA), polyetherimide (PEI), polyvinylidene fluoride (PVDF), polymethylmethacrylate (PMMA), polyether ether ketone (PEEK), polyether ketone (PEK), polyetherimide (PEI), polyethersulfone (PES), polyphenylene oxide (PPO), polystyrene (PS), polysulfone (PSU), polyvinyl chloride (PVC), liquid crystal polymer (LCP), thermoplastic elastomer (TPE), ultra-high molecular weight polyethylene (UHMWPE), or alloys of the above described thermoplastic materials, such as an alloy of acrylonitrile butadiene styrene (ABS) and polycarbonate (PC) or an alloy of acrylonitrile butadiene styrene (ABS) and polyamide (PA).

[0058] In other embodiments, the clamshell components **102a**, **102b** can be made of a metallic material that is bent, pressed, forged, machined, or otherwise processed to achieve a desired shape. The metallic clamshell can be made of the following metals: aluminum, titanium, magnesium, steel alloy, nickel alloy, titanium alloys, aluminum alloys, or any combination thereof.

[0059] In further embodiments, the clamshell components **102a**, **102b** can be made of a combination of multiple materials. For example, referring to FIG. 6, the clamshell components **102a**, **102b** can comprise multiple layers of metal **230a**, each separated from one another by a thermoplastic or thermoplastic composite layer **230b**. The metallic and thermoplastic materials used can be any of those listed above. In many embodiments, the metal and thermoplastic layers **230b** can be bonded by heating and pressing the layers together. Each of the thermoplastic layers **230b** may be a layup, made up of multiple layers. The illustrated embodiment comprises three layers of metal **230a** and two layers of composite **230b**. However, in other embodiments, the multi-layered clamshell components **102a**, **102b** can comprise 1-7 layers of metal **230a** and/or 1-7 layers of composite **230b**. In some embodiments, the clamshell components **102a**, **102b**

can comprise 1 layer of metal and multiple layers of composite, where the materials do not alternate, but the composite layers are stacked upon one another.

[0060] In some embodiments, a through-hole is provided to allow for air to flow through the paddle during a swing, thereby reducing drag, increasing swing speed and reduce muscle fatigue. The through-hole may be formed in a lower region of the head **101** or an upper region of the handle.

[0061] The thermoplastic composite material can include fillers, such as fibers, beads, or other structures comprising various materials (described below) that are mixed with the thermoplastic polymer. The fillers can provide structural reinforcement, weighting, lightening, or various other characteristics to the thermoplastic composite material. In many embodiments, the fillers can comprise carbon or glass. However, in other embodiments, the fillers can comprise other suitable materials. For example, the fillers can comprise aramid fibers (e.g., Nomex, Vectran, Kevlar, Twaron), bamboo fibers, natural fibers (e.g., cotton, hemp, flax), metal fibers (e.g., titanium, aluminum), glass beads, tungsten beads, or ceramic fibers (e.g., titanium dioxide, granite, silicon carbide).

[0062] In many embodiments, the thermoplastic composite material can comprise 30-40% fillers by volume. In other embodiments, the thermoplastic composite material can comprise up to 55%, up to 60%, up to 65%, or up to 70% fillers by volume.

[0063] The handle **104** can include injection molded components to simplify fabrication and reduce costs. For example, the handle **104** can include an injection-molded first handle member **104a** and an injection-molded second handle member **104b**, as shown in FIG. 5. The first handle member **104a** and the second handle member **104b** can be formed from the same mold. The handle members **104a**, **104b** can comprise exterior surfaces and define cavities.

[0064] The first handle member **104a** and the second handle member **104b** can be injection molded, thermoformed, integrally stamped with the respective clamshell component **102a**, **102b**, as illustrated, or otherwise bent and pressed into a desired shape. The first handle member **104a** and the second handle member **104b** can be adhered together through the use of mechanical fasteners, adhesives, ultrasonic welding, or any other suitable means. Multi-component handles can be manufactured with one or more identical pieces to reduce the need for additional tooling and increase ease of assembly.

A. Face Plates and Core Materials, Manufacturing, and Assembly

[0065] In some embodiments, the face plates **123a**, **123b** can comprise shapes, contours, or other structural bonding features that improve bonding to the core **134** or modulate response in targeted locations. For example, the face plates **123a**, **123b** can include grooves or raised embossing to increase surface area available for bonding, which can also promote even and controlled distribution of adhesive. The structural bonding features are on the face plate **122** interior surfaces. In other embodiments, the core **134** and the face plates **123a**, **123b** can be mechanically secured together. In further embodiments, both adhesive bonding and mechanical attachment means are used.

[0066] The first face plate **123a** and the second face plate **123b** can be made of any material, such as: metals, polymers (e.g. thermoplastic polyurethane, thermoplastic elastomer),

composites, or any combination thereof. In some embodiments, the face plate materials are chosen from a group consisting of: carbon fiber, fiberglass, or graphite. Metallic face plate materials include aluminum, titanium, magnesium, nickel alloy, titanium alloys, or aluminum alloys. In some embodiments, the face plates **123a**, **123b** can comprise a generally rectangular shape with curved corners. In many embodiments, the lower edges of the face plates **123a**, **123b**, nearest the handle **104a**, **104b**, are angled downward, such that they form an obtuse angle with the handle **104**.

III. EXAMPLES

Example 1—Manufacturing Time Comparison

[0067] Described herein is an exemplary embodiment of a clamshell pickleball paddle comprising a metallic material, hereafter referred to as “Exemplary Paddle 1.” Exemplary Paddle 1 is compared to a pickleball paddle constructed from a typical wet carbon fiber layup process, hereafter referred to as “Prior Art Paddle 1.” The manufacturing steps and time were measured and are compared in Table 1, below.

Step	Exemplary Paddle 1 Manufacturing Steps	Time (s)	Step	Prior Art Paddle 1 Manufacturing Steps	
1	Blanking	1	1	CF Laminate - Ply Cutting	300
2	Forming	5	2	CF Laminate - Layup	1000
3	Stamping	5	3	CF Laminate - Consolidation	2400
4	Blasting	10	4	Part Moving and Handling	120
5	Part Moving and Handling	120	5	Laminating CF Laminates to Core	1800
6	Foaming	1800	6	Trimming	30
7	Handle Sleeve Install	240	7	Cleaning/Drying/Masking	300
			8	Part Moving and Handling	120
			9	Paddle - Plasma Treat Surfaces	120
			10	Apply Epoxy	120
			11	Clamp and Cure	1800
			12	Grip Install	240
Total		2181	Total		8350

[0068] The manufacturing process to completely manufacture Exemplary Paddle 1 includes six main steps and one part handling step. The steps required to manufacture and assemble Exemplary Paddle 1 include: blanking, forming, and stamping the sheet metal, blasting the face plates **123a**, **123b** to create face texture, applying foam to the clamshell cavity, and installing the grip.

[0069] The manufacturing process to completely manufacture Prior Art Paddle 1 includes 10 main steps and two part handling steps. The steps required to manufacture and assembly Prior Art Paddle 1 include: cutting the carbon fiber (CF) laminate ply to shape, constructing the layup with multiple CF plies, joining the CF layup plies together (also called “consolidation”), laminating the layup to the core, trimming excess CF layup, cleaning, drying and masking the wet CF layup, plasma treating paddle surfaces for desired material characteristics, applying epoxy to the paddle through an automated process, and installing the grip.

[0070] Both the number of steps and the overall manufacturing time are reduced with the Exemplary Paddle 1. Referring to Table 1, the Exemplary Paddle 1 manufacturing process requires seven steps, while the Prior Art Paddle 1 manufacturing process requires 12 steps. The total time to

manufacture Exemplary Paddle 1 is about 74% less than the total time to manufacture Prior Art Paddle 1.

[0071] Similarly, when considering only the exterior of the head of the paddle, the time needed to manufacture the Exemplary Paddle 1 is significantly less than that needed to manufacture Prior Art Paddle 1. Initial forming of the metal material used for Exemplary Paddle 1 takes approximately 21 seconds and includes the following steps: blanking, forming, stamping, and blasting. Initial forming of the carbon fiber wet layup material used for Prior Art Paddle 1 includes the following steps: ply cutting, layup construction, and consolidation. Consequently, the time required for initial forming of the metal material of Exemplary Paddle 1 takes approximately 3700 seconds and is about 176 times less than the time required for initial forming of the carbon fiber material of Prior Art Paddle 1.

[0072] In conclusion, the process of manufacturing the metal clamshell of Exemplary Paddle 1 requires substantially less time and fewer steps when compared with the process of manufacturing a typical carbon fiber layup of Prior Art Paddle 1.

CLAUSES

[0073] Clause 1: A pickleball paddle comprising: a first clamshell component, including: a first head portion, a first rim portion surrounding the first head portion, a first handle portion, a second clamshell component, including: a second head portion, a second rim portion surrounding the second head portion, a second handle portion, wherein, when the first rim portion is joined to the second rim portion, the first and second clamshell components together define a cavity; and a core disposed in the cavity.

[0074] Clause 2: The pickleball paddle of claim 1, wherein the first rim portion comprises a first side wall, and the second rim portion comprises a second side wall.

[0075] Clause 3: The pickleball paddle of claim 2, wherein the first side wall is adhered to the second side wall.

[0076] Clause 4: The pickleball paddle of claim 1, wherein the first side wall is joined to the second side wall by a wall.

[0077] Clause 5: The pickleball paddle of claim 1, wherein each of the first and second clamshell components comprises a metal material.

[0078] Clause 6: The pickleball paddle of claim 5, wherein the metal material is aluminum, titanium, or an alloy of aluminum or titanium.

[0079] Clause 7: The pickleball paddle of claim 1, wherein the first and second handle portions define a handle cavity.

[0080] Clause 8: The pickleball paddle of claim 7, further comprising a handle fill disposed in the handle cavity.

[0081] Clause 9: The pickleball paddle of claim 1, wherein the core comprises a foam material.

[0082] Clause 10: The pickleball paddle of claim 9, wherein the foam material comprises an expanding foam.

[0083] Clause 11: The pickleball paddle of claim 1, further comprising a grip coupled to and surrounding the first and second handle portions.

[0084] Clause 12: The pickleball paddle of claim 1, further comprising an edge guard surrounding a majority of the head and covering a seam between the first rim portion and the second rim portion.

[0085] Replacement of one or more claimed elements constitutes reconstruction and not repair. Additionally, benefits, other advantages, and solutions to problems have been described with regard to specific embodiments. The benefits, advantages, solutions to problems, and any element or elements that may cause any benefit, advantage, or solution to occur or become more pronounced, however, are not to be construed as critical, required, or essential features or elements of any or all of the claims, unless such benefits, advantages, solutions, or elements are stated in such claim.

[0086] As the rules to pickleball may change from time to time (e.g., new regulations may be adopted or old rules may be eliminated or modified by pickleball standard organizations and/or governing bodies such as the United States Pickleball Association (USPA)), pickleball equipment related to the apparatus, methods, and articles of manufacture described herein may be conforming or non-conforming to the rules of pickleball at any particular time. Accordingly, pickleball equipment related to the apparatus, methods, and articles of manufacture described herein may be advertised, offered for sale, and/or sold as conforming or non-conforming pickleball equipment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

[0087] The above examples may be described in connection with a pickleball paddle. Alternatively, the apparatus, methods, and articles of manufacture described herein may be applicable to other types of sports equipment such as a tennis racquet, a badminton racquet, etc.

[0088] Moreover, embodiments and limitations disclosed herein are not dedicated to the public under the doctrine of dedication if the embodiments and/or limitations: (1) are not

expressly claimed in the claims; and (2) are or are potentially equivalents of express elements and/or limitations in the claims under the doctrine of equivalents.

1. A pickleball paddle comprising:

a first clamshell component, including:

a first head portion,

a first rim portion surrounding the first head portion,

a first handle portion,

a second clamshell component, including:

a second head portion,

a second rim portion surrounding the second head portion,

a second handle portion,

wherein, when the first rim portion is joined to the second rim portion, the first and second clamshell components together define a cavity; and

a core disposed in the cavity.

2. The pickleball paddle of claim 1, wherein the first rim portion comprises a first side wall, and the second rim portion comprises a second side wall.

3. The pickleball paddle of claim 2, wherein the first side wall is adhered to the second side wall.

4. The pickleball paddle of claim 1, wherein the first side wall is joined to the second side wall by a weld.

5. The pickleball paddle of claim 1, wherein each of the first and second clamshell components comprises a metal material.

6. The pickleball paddle of claim 5, wherein the metal material is aluminum, titanium, or an alloy of aluminum or titanium.

7. The pickleball paddle of claim 1, wherein the first and second handle portions define a handle cavity.

8. The pickleball paddle of claim 7, further comprising a handle fill disposed in the handle cavity.

9. The pickleball paddle of claim 1, wherein the core comprises a foam material.

10. The pickleball paddle of claim 9, wherein the foam material comprises an expanding foam.

11. The pickleball paddle of claim 1, further comprising a grip coupled to and surrounding the first and second handle portions.

12. The pickleball paddle of claim 1, further comprising an edge guard surrounding a majority of the head and covering a seam between the first rim portion and the second rim portion.

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